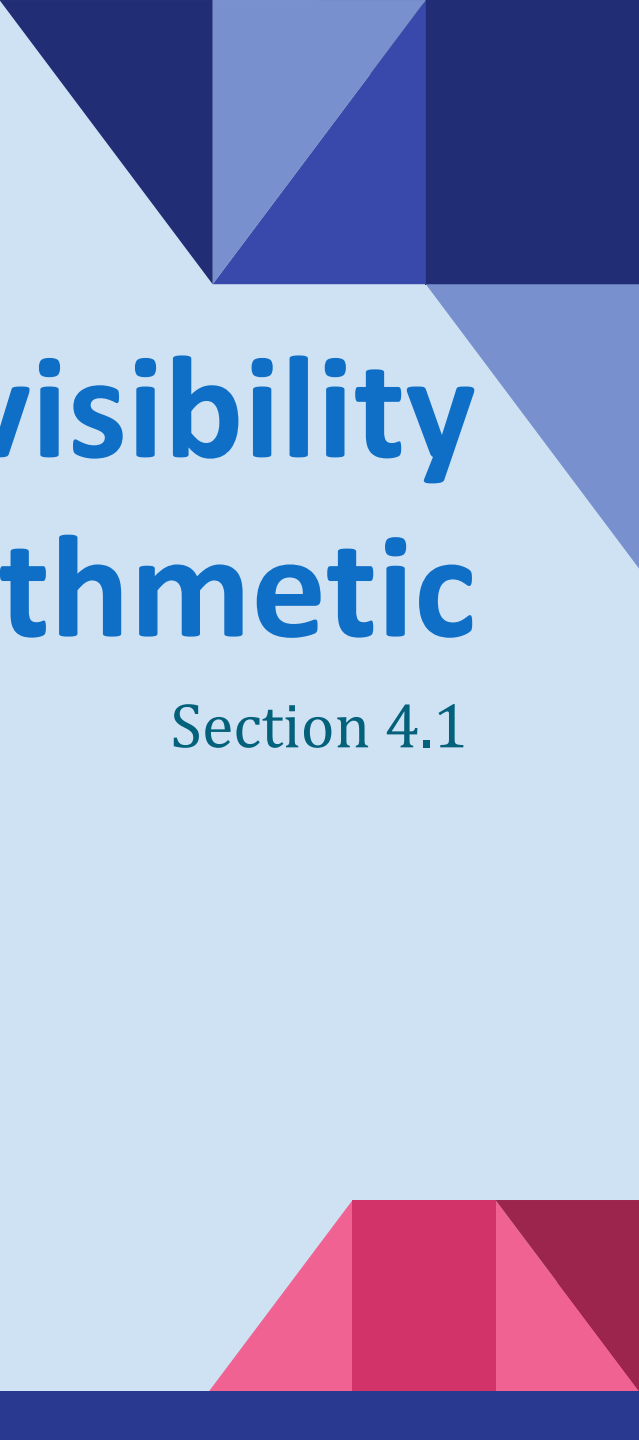


The slide features a light blue background. In the top right corner, there are several overlapping triangles in shades of dark blue and medium blue. In the bottom right corner, there are overlapping triangles in shades of red and pink. A solid dark blue horizontal bar runs across the bottom of the slide.

Lecture 10: Divisibility and Modular Arithmetic

Section 4.1

Section Summary

- Division
- Division Algorithm
- Modular Arithmetic

Division

Definition : If a and b are integers with $a \neq 0$, then a divides b if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor* or *divisor* of b and that b is a multiple of a .
- The notation $a \mid b$ denotes that a divides b .
- If $a \mid b$, then b/a is an integer.
- If a does not divide b , we write $a \nmid b$.

Example : Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Properties of Divisibility

Theorem 1 : Let a , b , and c be integers, where $a \neq 0$.

- i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- ii. If $a \mid b$, then $a \mid bc$ for all integers c ;
- iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t). \quad \text{Hence, } a \mid (b + c) \quad \blacktriangleleft$$

Corollary : If a , b , and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Division Algorithm

When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the “Division Algorithm,” but is really a theorem.

Division Algorithm : If a is an integer and d a positive integer, then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

- d is called the *divisor*.
- a is called the *dividend*.
- q is called the *quotient*.
- r is called the *remainder*.

Definitions of Functions

div and **mod**

$$q = a \text{ div } d$$

$$r = a \text{ mod } d$$

Division Algorithm

Since, $a = dq + r$ and $q = a \text{ div } d \Rightarrow q = \lfloor a/d \rfloor$,
 $r = a - dq \Rightarrow r = a - d * \lfloor a/d \rfloor$

Examples :

- What are the quotient and remainder when 101 is divided by 11?

Solution : The quotient when 101 is divided by 11 is

$9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.

- What are the quotient and remainder when -11 is divided by 3?

Solution : The quotient when -11 is divided by 3 is

$-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Division Algorithm

- Prove that if a is an integer that is not divisible by 3, then $(a+1)(a+2)$ is divisible by 3.

Solution : Assume that $3 \nmid a$.

So, we get a non-zero remainder, $r \in \{1, 2\}$ when a is divided by 3. In other words, $a = 3q + 1$ or, $a = 3q + 2$.

Where $q \in \mathbf{Z}$. Now,

Case 1: $a = 3q + 1$. Then, $(a+1)(a+2) = (3q+2)(3q+3) = 3(3q+2)(q+1)$. Therefore, $3 \mid (a+1)(a+2)$.

Case 2: $a = 3q + 2$. Then, $(a+1)(a+2) = (3q+3)(3q+4) = 3(q+1)(3q+4)$. Therefore, $3 \mid (a+1)(a+2)$.

Congruence Relation

Definition : If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus*.
- Two integers are congruent mod m if and only if they have the same remainder when divided by m .
- If a is not congruent to b modulo m , we write

$$a \not\equiv b \pmod{m}$$

Example : Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution :

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

More on Congruences

Theorem 4 : Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$.
- Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$. ◀

The Relationship between $(\text{mod } m)$ and $\text{mod } m$ Notations

- The use of “mod” in $a \equiv b \pmod{m}$ and $a \text{ mod } m = b$ are different.
 - $a \equiv b \pmod{m}$ is a **relation** on the set of integers.
 - In $a \text{ mod } m = b$, the notation **mod** denotes a **function**.
- The relationship between these notations is made clear in this theorem.

Theorem 3 : Let a and b be integers, and let m be a positive integer.

Then $a \equiv b \pmod{m}$ if and only if $a \text{ mod } m = b \text{ mod } m$.
(*Proof in the exercises*)

Congruences of Sums and Products

Theorem 5 : Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $b \cdot d = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. ◀

Example : Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Algebraic Manipulation of Congruences

- **Multiplying** both sides of a valid congruence by an integer preserves validity.

If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Adding** an integer to both sides of a valid congruence preserves validity.

If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Dividing** a congruence by an integer does **not** always produce a valid congruence.

Example : The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

Computing the **mod** m Function of Products and Sums

We use the following corollary to Theorem 5 to compute the **remainder of the product or sum of two integers** when divided by m **from the remainders** when each is divided by m .

Corollary : Let m be a positive integer and let a and b be integers. Then

$$(a + b) \text{ (mod } m) = ((a \text{ mod } m) + (b \text{ mod } m)) \text{ mod } m$$

$$\text{and } ab \text{ mod } m = ((a \text{ mod } m) (b \text{ mod } m)) \text{ mod } m.$$

(proof in textbook)